University of Tripoli

Faculty of Engineering

Department of Biomedical Engineering
Control and Instrumentation Division

Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease

A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Biomedical Engineering

Prepared by:

Suhail Mohamed Alhammali

ID: 2200208522

Supervised by:

Dr. Adel Adyaf

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Tripoli – Libya

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

﴿ وَقُل رَّبِّ زِدْنِي عِلْمًا ﴾

سورة طه، الآية 114

الإهداء

إلى والديّ، اللذين كان دعاؤهما وصبرهما سنداً لي في كل خطوة.

وإلى أساتذتي في قسم الهندسة الطبية الحيوية، الذين علّموني أن السؤال الصحيح أهمّ من الجواب السريع.

وإلى كل من يعمل على أن تصبح الرعاية الصحية أقرب متناولاً وأقلّ كلفة.

Declaration

I hereby declare that the project report entitled **Voice Signal Analysis Using Machine Learning for Early Detection of Parkinson's Disease**, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Biomedical Engineering from the Department of Biomedical Engineering at the University of Tripoli, is a bona fide work carried out by me under the supervision of **Dr. Adel Adyaf**.

This submission represents my ideas in my own words, and where the ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, ideas, facts, or sources in my submission. I understand that any violation of the above may result in disciplinary action by the institute and/or the university, and may also give rise to legal action by rights holders whose work has not been properly cited or for which permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma, or similar title at any other university.

Suhail Mohamed Alhammali

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First and foremost, all praise is due to Allah, whose guidance and blessing made the completion of this work possible.

I would like to express my sincere gratitude to my supervisor, **Dr. Adel Adyaf**, for his guidance, patience, and valuable feedback throughout every stage of this project. His insistence on scientific rigour shaped both the work and the way it is presented in this report.

I am also grateful to the academic and technical staff of the Department of Biomedical Engineering at the University of Tripoli for the knowledge and support they have provided throughout my years of study.

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Abstract

Parkinson's Disease impairs the neuromuscular control of speech, producing measurable changes in the voice that can appear early in the disease. This project develops and evaluates a complete, reproducible research prototype that analyses raw continuous-speech recordings for Parkinson's-related acoustic patterns using classical, explainable machine learning, with particular emphasis on honest validation.

From each recording, after standardized preprocessing and segmentation into 10-second chunks, 74 physiologically motivated acoustic features are extracted — fundamental-frequency statistics, jitter, shimmer, harmonics-to-noise ratio, smoothed cepstral peak prominence, pause statistics, and mel-frequency cepstral coefficients with their dynamics — and averaged per recording. Classifiers were compared on the MDVR-KCL corpus (37 subjects) under strictly subject-independent cross-validation, with model selection governed by rules declared before experimentation. The adopted Random Forest achieved a subject-level balanced accuracy of 0.822 ± 0.032, sensitivity of 0.771, specificity of 0.873, and an area under the receiver operating characteristic curve of 0.864, with pitch range and spectral variability as its leading features. A controlled comparison showed that a deliberately leaky chunk-level split inflates the same pipeline's result from 0.775 to 0.909, quantifying why subject-independent validation is essential. External validation on an independent Italian corpus (61 speakers), with the model frozen and recording bandwidth harmonized to neutralize a discovered sample-rate confound, yielded a speaker-level area under the curve of 0.701 and balanced accuracy of 0.629 across language and recording conditions. The accompanying screening prototype reports scores between 0.35 and 0.65 as inconclusive, attaches recording-condition warnings, and frames every output as a non-diagnostic research indication requiring professional clinical evaluation.

الملخص

يُضعف مرض باركنسون التحكم العصبي العضلي في الكلام، فيُحدث في الصوت تغيّرات قابلة للقياس قد تظهر مبكراً في مسار المرض. يقدّم هذا المشروع تطويراً وتقييماً لنموذج بحثي أولي متكامل وقابل لإعادة الإنتاج، يحلّل التسجيلات الصوتية الخام للكلام المتصل بحثاً عن الأنماط الصوتية المرتبطة بمرض باركنسون باستخدام تعلّم الآلة التقليدي القابل للتفسير، مع تركيز خاص على التحقّق الأمين من الأداء.

تُستخلص من كل تسجيل، بعد معالجة مسبقة موحّدة وتقطيع إلى مقاطع مدتها عشر ثوانٍ، أربع وسبعون خاصية صوتية ذات أساس فسيولوجي، هي: إحصاءات التردد الأساسي، والارتجاف، والتذبذب، ونسبة التوافقيات إلى الضجيج، وبروز القمة الكبسترالية المُنعَّم، وإحصاءات التوقّفات، ومعاملات الكبستروم على مقياس ميل مع ديناميكياتها؛ ثم يُؤخذ متوسطها لكل تسجيل. قورنت المصنّفات على قاعدة البيانات MDVR-KCL (سبعة وثلاثون شخصاً) بتحقّق متقاطع مستقل تماماً عن المتحدث، وخضع اختيار النموذج لقواعد أُعلنت قبل إجراء التجارب. حقّق نموذج الغابة العشوائية المعتمد دقة متوازنة على مستوى الشخص قدرها 0.822 بانحراف معياري 0.032، وحساسية 0.771، ونوعية 0.873، ومساحة تحت منحنى خصائص التشغيل للمستقبِل قدرها 0.864، وكان مدى التردد الأساسي وتغيّرية الطيف في مقدمة الخصائص المؤثرة. وأظهرت مقارنة مضبوطة أن تقسيماً خاطئاً متعمداً على مستوى المقاطع يضخّم نتيجة الأنبوب نفسه من 0.775 إلى 0.909، بما يقيس كمّياً ضرورة التحقّق المستقل عن المتحدث. كما أسفر التحقّق الخارجي على قاعدة بيانات إيطالية مستقلة (واحد وستون متحدثاً)، مع تجميد النموذج وتوحيد عرض النطاق الترددي لإبطال عامل مُربك مكتشف في معدّل العيّنات، عن مساحة تحت المنحنى قدرها 0.701 ودقة متوازنة قدرها 0.629 على مستوى المتحدث عبر اختلاف اللغة وظروف التسجيل. ويعرض النموذج الأولي المصاحب النتائج الواقعة بين 0.35 و0.65 بوصفها غير حاسمة، ويرفق تحذيرات عن جودة التسجيل، ويصوغ كل مخرجاته بوصفها مؤشراً بحثياً غير تشخيصي يستلزم تقييماً سريرياً متخصصاً.

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List of Abbreviations

Abbreviation	Definition
AUC	Area Under the Curve
CLI	Command-Line Interface
CPPS	Smoothed Cepstral Peak Prominence
CSV	Comma-Separated Values
CV	Cross-Validation
DC	Direct Current
F0	Fundamental Frequency
FN	False Negative
FP	False Positive
HC	Healthy Control
HNR	Harmonics-to-Noise Ratio
MDS	Movement Disorder Society
MDVR-KCL	Mobile Device Voice Recordings at King's College London
MFCC	Mel-Frequency Cepstral Coefficient
ML	Machine Learning
MLP	Multilayer Perceptron
PD	Parkinson's Disease
RBF	Radial Basis Function
RF	Random Forest
RMS	Root Mean Square
ROC	Receiver Operating Characteristic
SNR	Signal-to-Noise Ratio
SVM	Support Vector Machine
TN	True Negative

TP	True Positive
WAV	Waveform Audio File Format

Chapter 1

Introduction

1.1 Overview

This report documents the design, implementation, and evaluation of a research prototype that analyses raw voice recordings for acoustic patterns associated with Parkinson's Disease (PD) using classical, explainable Machine Learning (ML). The system covers the complete chain from an audio file to a cautious screening indication: signal preprocessing, acoustic feature extraction, model training, subject-independent validation, external validation on an independent dataset, and a file-based screening application. This chapter introduces the medical and engineering motivation behind the project, states the problem formally, lists the project objectives, defines the scope and the non-diagnostic framing of the system, and outlines the organization of the remainder of the report.

1.2 Background and Motivation

Parkinson's Disease is the second most common neurodegenerative disorder after Alzheimer's disease. It is caused by the progressive loss of dopamine-producing neurons in regions of the brain that control movement, and it manifests as tremor, muscular rigidity, slowness of movement, and reduced movement amplitude. The disease burden is large and growing: the Global Burden of Disease study estimated that about 6.1 million people were living with PD in 2016, more than double the number in 1990, a rise driven mainly by population ageing [1].

Despite this burden, no simple objective test for PD is available in routine practice. The diagnosis remains clinical: it is established from motor signs observed by a neurologist, following criteria such as those of the Movement Disorder Society (MDS) [2]. Post-mortem morphometric evidence indicates that a substantial proportion of the dopaminergic neurons of the substantia nigra is typically already lost by the time the characteristic motor signs appear [3], which narrows the window in which future disease-modifying treatments could be most useful. These facts together motivate the search for inexpensive, non-invasive measurements that could support earlier referral of at-risk individuals to clinical evaluation [2, 3].

The human voice is a promising carrier of such measurements. Speaking is a motor activity that depends on the coordinated action of the respiratory muscles, the vocal folds of the larynx, and the articulators of the vocal tract; the same neuromuscular impairments that affect limb movement in PD also affect these structures. Speech and voice impairment — reduced loudness, monotone pitch, breathiness, and imprecise articulation, collectively described as hypokinetic dysarthria — has been reported in approximately 90% of a large clinical sample of PD patients [4]. Importantly for screening, acoustic abnormalities appear early: quantitative acoustic analysis detected some form of vocal impairment in about 78% of patients with early, untreated PD [5], and a longitudinal case study found measurable reduction of fundamental-frequency variability in recordings made about five years before the clinical diagnosis [6].

A voice recording can be captured in seconds with a commodity microphone or smartphone, at negligible cost, with no discomfort, and repeatedly. If acoustic measurements of such recordings can be related to PD status with useful reliability, voice analysis could serve as a low-cost screening-support signal, flagging individuals for whom a clinical examination is advisable. This prospect has produced a large research literature and several public datasets, including the Mobile Device Voice Recordings at King's College London (MDVR-KCL) corpus used in this project [7].

Two considerations shaped the engineering approach taken here. First, in a biomedical setting, the ability to explain *why* a model produced its output is valued alongside raw performance; classical ML models operating on physiologically interpretable acoustic features were therefore preferred over deep learning, whose learned representations are difficult to interpret and whose data requirements far exceed the 37 subjects available.

Second, published results in this field frequently report accuracies above 95%, which, as examined in Chapters 2 and 4, often stems from evaluation protocols that let the same speaker appear in both training and test data. Producing an honest, subject-independent, and externally tested estimate of performance was therefore treated not as an obstacle but as a central contribution of the project.

1.3 Problem Statement

The problem addressed by this project is stated as follows: given a raw voice recording of continuous speech, determine — by means of reproducible signal processing and classical machine learning — whether the acoustic pattern of the recording is closer to that of a group of speakers with PD or to that of a group of healthy control speakers, and report the result with an honest, subject-independent estimate of reliability and with wording appropriate to a non-diagnostic research tool.

Solving this problem requires answering several subsidiary questions: which acoustic measurements meaningfully capture Parkinson's-related voice changes in continuous speech; how such measurements must be extracted so that training, evaluation, and prediction remain strictly consistent; how a model must be validated when several recordings originate from the same person; how much of the measured performance survives transfer to a different dataset, language, and recording channel; and how the output of such a system must be presented to avoid overstating what it knows.

1.4 Project Objectives

The objectives were divided into a primary set, which defines the minimum complete system, and an extended set, which was undertaken after the primary objectives were met.

1.4.1 Primary Objectives

1. Inspect and verify the MDVR-KCL dataset before any modelling: file integrity, class labels, subject identities, and the feasibility of subject-independent validation.

2. Implement a consistent preprocessing pipeline for raw audio (mono conversion, fixed sample rate, offset removal, silence trimming, and amplitude normalization) that deliberately avoids operations that would distort clinically meaningful signal irregularities.

3. Implement, as a core contribution, the extraction of physiologically motivated acoustic features: fundamental frequency statistics, jitter, shimmer, harmonics-to-noise ratio, and mel-frequency cepstral coefficient statistics.

4. Train and compare classical ML classifiers (logistic regression, support vector machine, random forest, and multilayer perceptron) against a majority-class baseline.

5. Evaluate all models with strictly subject-independent cross-validation, reporting balanced accuracy, sensitivity, specificity, F1 score, confusion matrices, and the area under the receiver operating characteristic curve, together with their variability across folds and random seeds.

6. Build a file-based screening prototype (browser application and command-line interface) that reuses the identical pipeline and presents results with mandatory non-diagnostic wording.

1.4.2 Extended Objectives

7. Improve classification performance through recording segmentation and additional continuous-speech features (smoothed cepstral peak prominence, pause statistics, and delta mel-frequency cepstral coefficients) under model-selection rules declared before experimentation, so that the improvement process itself cannot overfit the validation data.

8. Demonstrate quantitatively, on the project's own data, how an incorrect validation design inflates reported performance.

9. Validate the final, frozen model externally on an independent dataset recorded in a different language and under different conditions, and report the resulting generalization gap.

10. Extend the prototype with uncertainty-aware reporting: an inconclusive score band, recording-condition warnings, and a microphone demonstration mode explicitly framed as out-of-domain.

All ten objectives were achieved; the evidence is presented in Chapter 4.

1.5 Scope and Non-Diagnostic Framing

The system developed in this project is a research screening-support prototype. Throughout this report, and inside the software itself, its output is described as a non-diagnostic indication of similarity between the acoustic pattern of a recording and the patterns of the dataset groups. The system does not diagnose Parkinson's Disease, does not measure any person's medical risk, and must not replace evaluation by a qualified healthcare professional. Many conditions other than PD — among them common respiratory infections, ageing, smoking, and fatigue — also alter voice quality, and the datasets used are far too small and too homogeneous to support any clinical claim. The quantitative results presented in this report characterize performance over the datasets studied, not the accuracy of any statement about an individual. A dedicated discussion of limitations and the ethical constraints applied to the system's wording is given in Section 4.12.

Within these boundaries, the scope of the work comprises: the MDVR-KCL dataset as the training corpus; raw audio as the only input representation, with all features extracted by the project's own code; classical machine learning as the modelling approach; an independent Italian corpus as an external test set; and a file-based prototype with an additional microphone demonstration mode. Live clinical deployment, longitudinal patient tracking, and deep learning approaches are outside the scope of this project.

1.6 Report Organization

The remainder of this report is organized as follows.

Chapter 2 — Background and Literature Review presents the clinical background of PD and its effect on speech production, defines the acoustic measurements used in voice analysis, introduces the classical ML models employed, examines the data-leakage problem that affects much of the published literature, and reviews related work on the datasets used here.

Chapter 3 — Methodology specifies the complete system: the dataset inspection protocol, the preprocessing chain, recording segmentation, the 74-feature extraction stage, the classification pipeline, the subject-independent validation protocol, the pre-declared model-selection rules, the external validation design, and the prototype.

Chapter 4 — Results and Discussion reports the outcomes of every stage: dataset inspection, feature validation, baseline modelling, the model-improvement experiments, the final model, feature importance, the validation-design demonstration, external validation, and the prototype, followed by a general discussion and a dedicated section on limitations and ethical considerations.

Chapter 5 — Conclusion and Future Work summarizes the findings in points and lists recommendations for future development.

Four appendices provide the complete feature inventory, the full experimental results, a reproduction guide for the software repository, and key code excerpts.

Chapter 2

Background and Literature Review

2.1 Overview

This chapter provides the scientific foundation on which the project is built. Section 2.2 summarizes the clinical background of Parkinson's Disease (PD). Section 2.3 explains how the human voice is produced and how PD alters it. Section 2.4 defines the acoustic measurements used to quantify those alterations, together with the software tools that compute them. Section 2.5 introduces the classical Machine Learning (ML) models employed in this project. Section 2.6 examines the validation problem that affects much of the published literature in this field. Section 2.7 reviews related work and the public datasets used, and Section 2.8 states the research gap that this project addresses.

2.2 Parkinson's Disease

PD is a chronic, progressive neurodegenerative disorder caused by the loss of dopamine-producing neurons, predominantly in the substantia nigra pars compacta, a region of the midbrain involved in the control of movement. Post-mortem morphometric analysis indicates that this loss is regionally selective and that a substantial proportion of the affected neurons has already degenerated by the time motor symptoms appear [3]. The cardinal motor manifestations are bradykinesia (slowness of movement), muscular rigidity, rest tremor, and, in later stages, postural instability; the disease also produces a wide range of non-motor symptoms [2].

The burden of the disease is substantial and increasing. The Global Burden of Disease study estimated that about 6.1 million individuals were living with PD worldwide in 2016, compared with 2.5 million in 1990, an increase driven mainly by the growing number of older people [1]. There is at present no cure and no routine laboratory test; the diagnosis is clinical, based on the motor examination and supporting criteria formalized by the Movement Disorder Society (MDS) [2].

Two aspects of this clinical picture motivate the present project. First, because the diagnosis depends on visible motor signs, it is typically made only after the underlying neuronal loss is well advanced [3]. Second, the muscles of respiration, the larynx, and the articulators are affected by the same motor impairment as the limbs, so measurable changes in the voice can accompany — and in some reported cases precede — the visible signs [5, 6]. An inexpensive measurement channel that reflects early motor impairment is therefore of practical interest as a screening-support signal, provided its output is interpreted cautiously and never treated as a diagnosis.

2.3 Speech Production and Hypokinetic Dysarthria

2.3.1 The Source–Filter Model of Voice Production

Human voice production is conventionally described by the source–filter model, illustrated in Figure 2.1. Three stages cooperate:

1. **Power source.** The lungs supply a steady stream of pressurized air through the trachea.

2. **Sound source.** The two vocal folds of the larynx are brought together across the airway. Subglottal pressure forces them apart, and their elasticity, together with aerodynamic forces, snaps them shut again. This cycle repeats quasi-periodically, releasing a train of air puffs. The repetition rate of this cycle is perceived as the pitch of the voice and is called the Fundamental Frequency (F0); typical adult values lie roughly between 85 and 180 Hz for men and 165 and 255 Hz for women.

3. **Filter.** The pharynx, oral cavity, tongue, teeth, and lips form a resonating tube whose shape changes continuously during speech. This vocal-tract filter emphasizes

certain frequency regions and attenuates others, shaping the buzz of the source into distinguishable vowels and consonants.

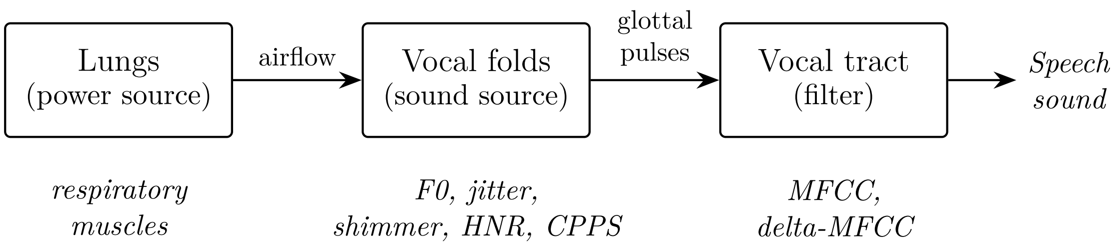

Figure (2.1): The source–filter model of voice production. The lower row indicates which acoustic feature families used in this project characterize each stage.

Two properties of this system matter for the measurements that follow. The rate of vocal-fold vibration determines F0, and because typical F0 ranges differ between adult men and women, absolute pitch values partly encode the speaker's sex — a potential confounder examined explicitly in Chapter 4. The *regularity* of the vibration determines perceived voice quality: in a healthy voice, successive cycles are nearly identical, while irregular timing or amplitude is heard as roughness, and incomplete closure of the folds is heard as breathiness.

2.3.2 Hypokinetic Dysarthria in Parkinson's Disease

Dysarthria denotes a group of speech disorders caused by impaired neuromuscular control of the speech apparatus. In their classical perceptual study, Darley, Aronson, and Brown identified distinct dysarthria patterns corresponding to different neurological lesions and associated parkinsonism with *hypokinetic* dysarthria [8]. Its perceptual hallmarks are reduced loudness, monotone pitch (reduced F0 variability), breathy or hoarse voice quality, imprecise consonant articulation, and disturbed speech timing.

These perceptual signs have measurable acoustic correlates, and their prevalence is high. In a sample of 200 PD patients, speech or voice impairment was present in approximately 90% [4]. Quantitative acoustic analysis of 46 early, untreated PD patients found some form of vocal impairment in about 78%, with measures of F0 variability among the most discriminative [5]. Timing is affected as well: compared with healthy speakers reading the same text, PD patients showed altered pause behaviour and a tendency to accelerate their speech rate [9]. A longitudinal case study further suggested that reduced

F0 variability can be detected in recordings made about five years before clinical diagnosis [6].

These findings define the measurement targets for this project: pitch variability, cycle-to-cycle regularity, the balance of periodic and noise energy, articulatory spectral dynamics, and pause behaviour.

2.4 Acoustic Measurement of the Voice

This section defines the acoustic quantities used in the project. All perturbation and cepstral measures are computed with the Praat speech analysis system [10] through its Python interface Parselmouth [11]; spectral features are computed with the librosa library [12].

2.4.1 Fundamental Frequency Statistics

F0 is estimated over short overlapping analysis frames using the autocorrelation method of Boersma [13], which locates the dominant periodicity of the waveform within a configured search range (75–500 Hz in this project, covering adult male and female voices). Frames without detectable periodicity (silences, unvoiced consonants) are excluded. From the resulting F0 track, summary statistics are computed: mean, median, standard deviation, minimum, maximum, range, and the voiced fraction (the proportion of frames in which voicing was detected). Given the monotone character of hypokinetic dysarthria, the dispersion statistics (standard deviation and range) are the most clinically meaningful [5, 6].

2.4.2 Jitter

Jitter quantifies the cycle-to-cycle instability of vocal-fold *timing*. Let T_i denote the duration of the i-th glottal period and N the number of periods. Local jitter is the mean absolute difference between consecutive periods, normalized by the mean period:

$$\text{jitter}_{\text{local}} = \frac{\dfrac{1}{N-1}\displaystyle\sum_{i=1}^{N-1} |T_i - T_{i+1}|}{\dfrac{1}{N}\displaystyle\sum_{i=1}^{N} T_i}. \tag{2.1}$$

Variants used in this project average over small neighbourhoods of periods to reduce sensitivity to isolated measurement errors: absolute local jitter (the numerator of Eq. 2.1, in seconds), the Relative Average Perturbation (RAP, three-period neighbourhood), and the five-period Perturbation Quotient (PPQ5). In healthy sustained phonation, local jitter is typically below about 1%; impaired neuromuscular control raises it.

2.4.3 Shimmer

Shimmer applies the same idea to cycle *amplitudes* A_i:

$$\text{shimmer}_{\text{local}} = \frac{\dfrac{1}{N-1}\displaystyle\sum_{i=1}^{N-1} |A_i - A_{i+1}|}{\dfrac{1}{N}\displaystyle\sum_{i=1}^{N} A_i}. \tag{2.2}$$

The project uses local shimmer, its decibel form, and the three-, five-, and eleven-period Amplitude Perturbation Quotients (APQ3, APQ5, APQ11). Weak or inconsistent vocal-fold closure produces uneven pulse strength and therefore raised shimmer. Because shimmer is normalized by the mean amplitude, a uniform change of recording gain does not affect it.

2.4.4 Harmonics-to-Noise Ratio

The voice signal is a mixture of a periodic component (the harmonics of F0) and an aperiodic noise component caused by turbulent airflow. The Harmonics-to-Noise Ratio (HNR) expresses their energy ratio in decibels. Using the normalized autocorrelation approach of Boersma, if $r_{\max}$ is the height of the normalized autocorrelation peak at the detected period (the fraction of signal power that is periodic), then

$$\mathrm{HNR} = 10 \log_{10} \frac{r_{\max}}{1 - r_{\max}} \text{ dB.} \tag{2.3}$$

A clear voice may exceed 20 dB, while breathy voices score much lower [13]. Incomplete glottal closure — a recognized feature of hypokinetic dysarthria — lowers HNR.

2.4.5 Smoothed Cepstral Peak Prominence

Jitter, shimmer, and HNR presuppose that individual glottal cycles can be identified reliably, which holds for sustained vowels but only approximately for continuous speech. The Smoothed Cepstral Peak Prominence (CPPS) avoids this requirement. The cepstrum is obtained by taking a second spectral transform of the logarithmic power spectrum; a strongly periodic voice, whose spectrum contains regularly spaced harmonics, produces a sharp cepstral peak at the quefrency of the glottal period. CPPS measures how far this peak rises above the regression line fitted to the smoothed cepstrum, in decibels. A meta-analysis of acoustic correlates of perceived voice quality found cepstral peak prominence measures to be among the strongest and most robust correlates of overall dysphonia, in both sustained vowels and continuous speech [14]. Breathier, noisier voices produce weaker cepstral peaks and therefore lower CPPS.

2.4.6 Pause and Timing Statistics

Speech timing is quantified here with an energy-based segmentation of the recording into speech stretches and silences. Gaps of at least 0.2 s are counted as pauses, from which four features are derived: pauses per minute, mean pause duration, the fraction of total time spent in pauses, and the mean duration of continuous speech stretches. The clinical motivation follows from the altered pause behaviour documented in PD reading tasks [9]. These features have the additional practical advantage of depending on timing rather than on spectral content, which makes them comparatively robust to changes of microphone and recording channel.

2.4.7 Mel-Frequency Cepstral Coefficients and Their Dynamics

The Mel-Frequency Cepstral Coefficients (MFCC) provide a compact description of the short-term spectral envelope — effectively, of the vocal-tract filter configuration. The signal is divided into short frames; for each frame the power spectrum is computed, integrated into overlapping bands spaced on the mel scale,

$$m(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right), \tag{2.4}$$

which approximates the frequency resolution of human hearing; the band energies are logarithmized; and a discrete cosine transform yields the coefficients, of which the first 13 are retained here, following the representation introduced by Davis and Mermelstein [15]. Delta-MFCC coefficients are the frame-to-frame time derivatives of the MFCCs and describe how quickly the spectral envelope changes — a proxy for articulatory movement speed. In this project each coefficient trajectory is summarized by its mean and standard deviation over the analysed audio, yielding 26 MFCC and 26 delta-MFCC features. Because MFCCs describe the spectrum in detail, they are also sensitive to the microphone, room, and language — a property with consequences examined in the external validation (Chapter 4).

2.5 Classical Machine Learning Models

Four classifier families, all implemented in the scikit-learn library [16], are used in this project, together with a majority-class baseline that always predicts the more frequent class and therefore represents zero learned knowledge.

Logistic regression models the probability of class membership as a logistic function of a weighted sum of the features. It produces a linear decision boundary, and the learned weights indicate each feature's direction and strength of influence, making it the most directly interpretable model considered.

The Support Vector Machine (SVM) [17] seeks the decision boundary that separates the classes with the widest possible margin, which favours good generalization from limited data. With the Radial Basis Function (RBF) kernel, the data are implicitly mapped into a higher-dimensional space, allowing the boundary to be curved while the margin principle is retained.

The Random Forest (RF) [18] is an ensemble of decision trees, each trained on a bootstrap sample of the training data with random feature subsets considered at each split; the forest predicts by aggregating the votes of its trees. RFs handle features of mixed scales and nonlinear interactions naturally, resist overfitting through averaging, and provide a ranking of feature importance — a property used in Chapter 4 to interpret the final model.

The Multilayer Perceptron (MLP) is a small feed-forward neural network trained by gradient descent. It is included for completeness as the most flexible of the four families, but flexibility carries risk with only 37 subjects, and its behaviour under a confounder ablation (Chapter 4) illustrates exactly this danger.

The rationale for restricting the project to such classical models rather than deep learning is threefold: the sample size (37 subjects) is far below what deep networks require; the biomedical setting demands interpretable evidence of *what* the model uses; and the project's scientific focus is on honest validation, which is clearer to demonstrate with simple, well-understood models.

2.6 Validation Methodology and the Data-Leakage Problem

The value of any reported accuracy depends entirely on how the model was tested. When a dataset contains several recordings — or several excerpts of one recording — from the same person, a random split of those items between training and test sets allows the model to encounter each test speaker during training. The model can then succeed by *recognizing the speaker or the recording*, rather than the disease, and the measured performance no longer predicts behaviour on new people. This phenomenon is a form of data leakage, and in the clinical machine learning literature the distinction is drawn between *record-wise* and *subject-wise* cross-validation: Saeb et al. demonstrated that record-wise validation can dramatically overestimate diagnostic accuracy and argued that the validation scheme must approximate the intended use case — diagnosing *new* individuals [19].

The correct procedure is grouped, subject-independent validation: all data from one person is assigned as a block to either the training or the test side of every split. Where class proportions should also be preserved across folds, stratified grouped K-fold splitting is used. This project enforces subject independence mechanically — a runtime assertion halts execution if any subject ever appears on both sides of a split — and additionally demonstrates the size of the leakage effect empirically on its own data, by evaluating the same pipeline under a deliberately wrong chunk-level split (Chapter 4).

A related, subtler leakage channel concerns preprocessing statistics: if imputation values or feature scaling parameters are estimated on the full dataset before splitting, information about the test data enters training. The methodology of Chapter 3 therefore fits all such operations inside each training fold only.

2.7 Related Work and Datasets

Research on voice-based PD assessment spans two broad recording paradigms. The first uses *sustained vowels*, in which the speaker phonates a steady "ah"; this isolates phonatory function and suits the classical perturbation measures. Little et al. combined dysphonia measures of sustained vowels with an SVM classifier to separate PD speakers from controls [20], and Tsanas et al. extended this line with a set of 132 dysphonia measures and feature selection, reporting almost 99% classification accuracy on sustained-vowel data [21]. Results of this magnitude, here as elsewhere in the field, must be read together with the validation-design question discussed in Section 2.6: when several samples of the same speaker can fall on both sides of a random split, part of the measured performance may reflect speaker recognition rather than disease detection [19]. The second paradigm uses *connected speech* (read text or spontaneous conversation), which engages articulation and prosody in addition to phonation and is closer to natural speaking conditions [5, 9]; it is the paradigm of the present project.

Two public connected-speech datasets are used in this work. The MDVR-KCL corpus [7] contains smartphone recordings of read text and spontaneous dialogue from 37 English-speaking participants (21 healthy controls and 16 PD patients) and serves as the training corpus; its detailed inspection appears in Chapter 4. The Italian Parkinson's Voice and Speech database, collected by Dimauro et al. in the context of speech-intelligibility assessment, contains recordings of 65 Italian speakers in young healthy, elderly healthy, and PD groups [22]. The publicly mirrored copy of this database used in the present work comprises 61 of those speakers (15 young healthy controls, 22 elderly healthy controls, and 24 PD patients; Chapter 4), and serves here exclusively as an external, never-trained-on test set.

Published classification studies on MDVR-KCL and similar corpora frequently report accuracies above 95%. In light of the record-wise leakage mechanism described above [19], and of the demonstration in Chapter 4 that a chunk-level random split inflates this project's own balanced accuracy from 0.775 to 0.909 without any change to the model, such figures should be interpreted with caution unless subject-independent validation is stated explicitly.

2.8 Research Gap and Project Contribution

From the literature reviewed above, three gaps emerge. First, many reported results rest on validation schemes that do not separate speakers, leaving their practical meaning uncertain [19]. Second, cross-dataset evaluation — testing a trained system on an entirely different corpus, language, and recording channel — is rare, although it is the most direct measure of the generalization that a screening application would require. Third, reported systems seldom address how their output should be presented given the uncertainty involved; a bare class label overstates what a small-corpus model knows.

This project addresses all three gaps within an undergraduate scope: it enforces subject-independent validation mechanically and quantifies the leakage effect on its own data; it performs an unadapted external validation on a second corpus in a different language; and it delivers a prototype whose output includes an explicit inconclusive band, recording-condition warnings, and mandatory non-diagnostic wording. The complete methodology follows in Chapter 3.

Chapter 3

Methodology

3.1 Overview

This chapter specifies the complete system, from the raw audio file to the cautious screening indication, in the order in which data flows through it. Two principles govern every design decision and are worth stating before the details. The first is the *same-pipeline rule*: the identical preprocessing, segmentation, and feature-extraction code is used for training, for evaluation, and for predicting on a new file, and this identity is enforced mechanically rather than by convention. The second is *subject independence*: no data derived from a given person is ever allowed to influence a model that is evaluated on that person, and this too is enforced by the code itself.

Sections 3.2–3.6 describe the data path: system architecture, the dataset and its inspection, preprocessing, segmentation, and feature extraction. Sections 3.7–3.9 describe the modelling side: the classification pipeline, the validation protocol, and the pre-declared model-selection rules. Section 3.10 specifies the external validation design, Section 3.11 the prototype, and Section 3.12 the software environment and reproducibility measures.

3.2 System Architecture

Figure 3.1 shows the complete processing chain. A recording is first conditioned by a five-step preprocessing stage, then divided into 10-second chunks. From every chunk, 74 acoustic features are extracted; the recording is represented by the per-feature mean over its chunks. This vector enters a classification pipeline consisting of a median imputer, a standard scaler, and the classifier, which outputs a score between 0 and 1 for the PD class. Finally, an interpretation stage maps the score to one of three outcomes — closer to the Healthy Control (HC) class, closer to the PD class, or inconclusive — and attaches any recording-condition warnings and the mandatory non-diagnostic wording.

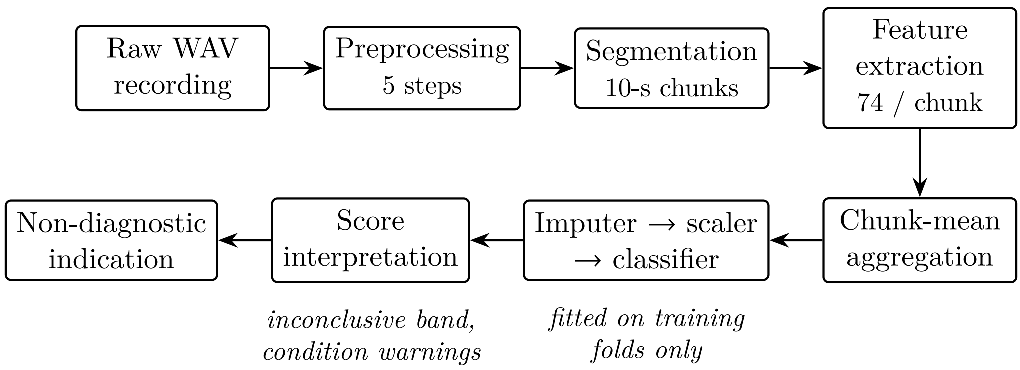

Figure (3.1): The complete processing chain. The same implementation of every stage is shared by training, evaluation, and prediction.

All numerical settings of every stage — sample rate, trimming threshold, pitch search range, segmentation lengths, spectral analysis parameters — are defined once in a single configuration module. When a model is trained, the full configuration, including the complete ordered list of feature names, is saved beside the model as a machine-readable contract. Before *every* prediction, the software compares this saved contract with the configuration of the running code; any difference, or a missing contract file, aborts the prediction with an instruction to retrain. The same-pipeline rule is therefore a verified property of the system, not a development convention.

3.3 Dataset and Inspection Protocol

3.3.1 The MDVR-KCL Corpus

The training corpus is the MDVR-KCL database [7], comprising smartphone voice recordings of English-speaking participants performing two tasks: reading a fixed text (ReadText) and holding a spontaneous conversation (SpontaneousDialogue). Table 3.1 summarizes its composition as verified by the inspection stage. All 73 recordings are mono WAV files sampled at 44100 Hz, with durations between 73.0 and 220.6 seconds (median 141.3 s). Each participant contributed at most one recording per task; 36 of the 37 subjects have exactly two recordings, and one PD subject has one.

Table (3.1): Composition of the MDVR-KCL corpus as used in this project.

Group	Subjects	ReadText	SpontaneousDialogue
Healthy Control (HC)	21	21	21
Parkinson's Disease (PD)	16	16	15
Total	37	37	36

3.3.2 Labels and Subject Identities

The class label of each file is determined from two independent sources: the folder in which the file resides (HC or PD) and a tag embedded in its filename. The two sources are cross-checked, and a file whose sources disagree would be excluded and reported as a label conflict rather than assigned silently; no such conflict occurred. The subject identity is parsed from the leading identifier token of the filename. This identity is the basis of subject-independent validation, so files without a recognizable subject identifier would likewise be excluded and reported; none occurred.

3.3.3 Integrity Checks

Before any modelling, an automated inspection verified: readability of every file; absence of empty, silent, or very short recordings; absence of duplicate audio content (by MD5 hash comparison, which protects against the same recording entering both training and test data under different names); label consistency; and absence of subjects appearing in more than one class. All 73 files passed every check. The inspection also recorded the dataset's known risks — small sample size, homogeneous recording conditions, class imbalance (42 HC versus 31 PD recordings), and the absence of per-subject demographic metadata — which shape the validation and analysis decisions described below.

3.4 Audio Preprocessing

Each recording passes through five conditioning steps, in fixed order:

1. **Mono conversion.** Multi-channel input is averaged to a single channel.

2. **Resampling to 44100 Hz.** All analysis assumes one fixed sample rate; 44100 Hz is the native rate of the corpus, so no resampling artefacts are introduced for the training data.

3. **DC-offset removal.** The signal mean is subtracted; a constant offset would bias amplitude-based measures such as shimmer.

4. **Edge silence trimming.** Leading and trailing segments more than 30 dB below the signal peak are removed. Internal pauses are deliberately retained, both because the pause features measure them and because cutting inside the recording would create artificial amplitude discontinuities.

5. **Peak normalization.** The signal is scaled so that its maximum absolute amplitude is 0.95, making recordings comparable across input gains. Because jitter, shimmer, and the other perturbation measures are relative quantities (Eqs. 2.1–2.2), a uniform gain change does not distort them.

No denoising, filtering, or spectral enhancement of any kind is applied. This is a deliberate scientific decision: noise-reduction algorithms operate precisely by detecting and suppressing signal irregularities, and cycle-to-cycle irregularity is exactly the clinical information that jitter, shimmer, HNR, and CPPS are designed to measure. Aggressive cleaning would make a disordered voice appear artificially healthy.

3.5 Segmentation and Aggregation

After preprocessing, each recording is divided into consecutive non-overlapping chunks of 10 seconds; a trailing chunk shorter than 5 seconds is discarded, and a recording shorter than the chunk length is processed as a single chunk. Applied to the corpus, this produced 1001 chunks (between 7 and 22 per recording).

Features are extracted per chunk, and the recording-level feature vector is the per-feature arithmetic mean over its K chunks:

$$\bar{x}_j = \frac{1}{K} \sum_{k=1}^{K} x_{j,k}, \qquad j = 1, \ldots, 74, \tag{3.1}$$

where $x_{j,k}$ is feature j measured on chunk k. Averaging many short-window measurements is statistically more stable than one measurement over a whole recording, and the experiments of Chapter 4 show that this representation improves classification. The chunks inherit the recording's subject identity, so the grouping required for honest validation is preserved — and Section 3.8 shows what happens when it is not.

3.6 Feature Extraction

Table 3.2 lists the eight feature families; their definitions and physiological meaning were given in Section 2.4. The complete inventory of all 74 names appears in Appendix A.

Table (3.2): The 74 acoustic features extracted from every chunk.

Family	Count	Tool
F0 statistics	7	Praat via Parselmouth
Jitter variants	4	Praat via Parselmouth
Shimmer variants	5	Praat via Parselmouth
Harmonics-to-noise ratio	1	Praat via Parselmouth
Smoothed cepstral peak prominence	1	Praat via Parselmouth
Pause and timing statistics	4	energy-based segmentation
MFCC summary statistics	26	librosa
Delta-MFCC summary statistics	26	librosa
Total	74	

The principal analysis settings are as follows. Pitch tracking and the perturbation measures use the autocorrelation method [13] with a search range of 75–500 Hz. CPPS uses Praat's power-cepstrogram analysis with a 60 Hz pitch floor, a 2 ms time step, and a peak search range of 60–330 Hz. Pause statistics use an energy threshold of 25 dB below peak with a minimum pause duration of 0.2 s. MFCC analysis uses 13 coefficients over 2048-sample windows with a 512-sample hop (approximately 46 ms and 12 ms at 44100 Hz), summarized by mean and standard deviation per coefficient, and likewise for their time derivatives [12, 15].

On chunks containing too little stable voicing, Praat cannot compute some perturbation measures; these values are recorded as missing rather than fabricated. In the corpus this affected at most 16 of the 1001 chunks for any single feature. Missing values are filled by the median imputer of the classification pipeline, which — critically — is fitted inside each training fold only (Section 3.7).

The extraction stage always computes all 74 features through a single code path; the original 43-feature set of the baseline phase (Chapter 4) is a named subset of the same table, so base and extended experiments differ only in column selection, never in extraction code.

3.7 Classification Pipeline

Every classifier is embedded in a three-stage pipeline executed as one object:

$$\text{median imputer} \quad \rightarrow \quad \text{standard scaler} \quad \rightarrow \quad \text{classifier}$$

The imputer replaces missing values with the median of the corresponding feature. The scaler standardizes each feature to zero mean and unit variance,

$$z_j = \frac{x_j - \mu_j}{\sigma_j}, \tag{3.2}$$

which is necessary because the features span incommensurate scales (mean F0 is of order 10^2 Hz while local jitter is of order 10^{-2}). Both stages estimate their statistics (μ_j, σ_j, medians) *from the training portion of each validation fold only*; embedding them in the pipeline object guarantees this and thereby closes the preprocessing leakage channel described in Section 2.6.

Five models are trained and compared, all implemented in scikit-learn [16]: a majority-class baseline; logistic regression (maximum 5000 iterations); an SVM with RBF kernel [17]; a Random Forest of 300 trees with unrestricted depth [18]; and an MLP with hidden layers of 32 and 16 units. All models except the baseline use balanced class weighting, which scales each training example inversely to its class frequency so that the minority PD class is not systematically under-predicted. A fixed random seed makes every training run reproducible.

3.8 Subject-Independent Validation Protocol

3.8.1 Grouped Cross-Validation

Evaluation uses stratified grouped 5-fold cross-validation, with the subject identity as the grouping variable, repeated with three random seeds (42, 7, and 2025). The grouped splitter assigns all recordings (and therefore all chunks) of one subject to the same side of every split, while stratification keeps the HC/PD proportion approximately constant across folds. Figure 3.2 illustrates the principle. Within each of the 3×5 evaluations, the model is fitted on the training subjects and produces predictions for the held-out subjects; collecting these across folds yields exactly one *out-of-fold* prediction per recording per seed, made in every case by a model that never saw that person.

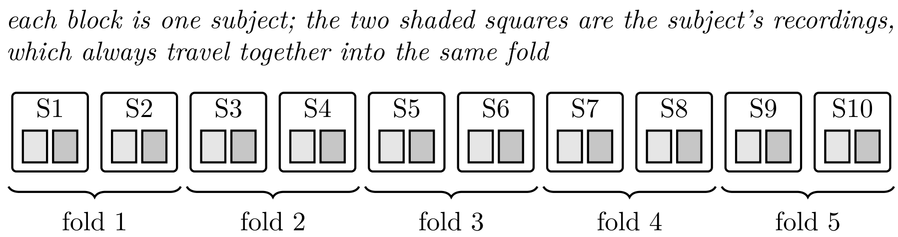

Figure (3.2): Subject-grouped cross-validation (illustrated with 10 subjects). A subject's recordings are never divided between training and test folds.

The splitter's guarantee is not taken on trust. Inside every fold, the implementation recomputes the intersection of training and test subjects and halts the program if it is non-empty:

```
train_subjects = set(groups.iloc[train_idx])
test_subjects = set(groups.iloc[test_idx])
overlap = train_subjects & test_subjects
if overlap:
    raise RuntimeError(
        f"LEAKAGE: subjects {overlap} appear in train AND test"
    )
```

This assertion never triggered in any experiment of this project.

Results are reported at two levels. The *recording level* scores each audio file independently. The *subject level* averages the out-of-fold PD scores of a subject's recordings and applies the 0.5 threshold to the mean; it is the clinically more relevant level, since the practical question concerns persons, not files. All headline metrics are stated as mean ± standard deviation over the three seeds.

3.8.2 Metrics

With TP, TN, FP, and FN denoting true positives (PD correctly identified), true negatives, false positives, and false negatives, the reported metrics are:

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \tag{3.3}$$

$$\text{balanced accuracy} = \tfrac{1}{2}\left(\text{sensitivity} + \text{specificity}\right), \tag{3.4}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad F_1 = \frac{2\,(\text{precision} \times \text{sensitivity})}{\text{precision} + \text{sensitivity}}. \tag{3.5}$$

Balanced accuracy is the primary metric because the classes are imbalanced: a baseline that always answers HC attains $42/73 \approx 0.575$ plain accuracy while detecting no patient at all, but only 0.5 balanced accuracy, which correctly exposes it. Confusion matrices are reported for all models. In addition, the Area Under the Receiver Operating Characteristic Curve (AUC) summarizes the model's scores independently of the 0.5 threshold; it equals the probability that a randomly chosen PD case receives a higher score than a randomly chosen HC case, with 0.5 corresponding to chance and 1.0 to perfect ranking.

Finally, a pre-declared warning rule guards against too-good-to-be-true results: any subject-independent accuracy, balanced accuracy, or AUC reaching 0.95 triggers an explicit investigation for leakage, duplicates, or confounding before the result may be reported as a success.

3.9 Pre-Declared Model-Selection Protocol

The accuracy-improvement study of Chapter 4 compares data representations and models. Because repeatedly evaluating variants on the same validation data can itself overfit — the selection, rather than the model, adapts to the data — the following rules were fixed *before* any experiment was run:

1. All variants are evaluated with the identical protocol of Section 3.8 (same folds, same three seeds).

2. The primary metric is subject-level balanced accuracy, averaged over seeds.

3. A more complex variant is adopted only if it exceeds the simpler alternative by more than +0.02 on the primary metric; otherwise the simpler variant is retained. The margin is one-sided — it applies to improvements — and its size reflects the variability observed at this sample size: a large fold-to-fold spread (Appendix B) and a resulting seed-to-seed variability of the mean of about ±0.03, below which differences are indistinguishable from noise.

4. Any result at or above 0.95 halts the study for a leakage investigation (Section 3.8.2).

5. Every evaluated configuration is reported, including rejected ones.

Table 3.3 lists the seven data-representation variants evaluated under these rules. Hyperparameter tuning (variant V4) is performed by *nested* grouped search: within each outer training fold, an inner 3-fold grouped cross-validation selects, by balanced accuracy, the SVM cost and kernel width ($C \in \{0.1, 1, 10, 100\}$, $\gamma \in \{\text{scale}, 0.01, 0.001\}$), the RF depth and feature fraction (depth $\in \{\text{unrestricted}, 5, 10\}$, features per split $\in \{\sqrt{p}, 0.3p\}$), and the logistic-regression regularization ($C \in \{0.01, 0.1, 1, 10\}$); the outer test subjects play no role in the choice. The ensemble (V5) soft-votes the per-fold-tuned SVM and RF.

Table (3.3): Data-representation variants of the model-selection study.

Variant	Representation	Features
V0	whole-recording features (baseline phase)	43
V1	chunk features averaged per recording (Eq. 3.1)	74
V2	chunk-level classification, scores averaged	43
V3	chunk-level classification, scores averaged	74
V4	best of V0–V3 with nested hyperparameter tuning	—
V5	tuned SVM + RF soft-voting ensemble	—
V6	chunks summarized by mean and standard deviation	148

3.10 External Validation Design

The final model is additionally evaluated on the Italian Parkinson's Voice and Speech database [22], under a design agreed before the evaluation was run:

1. **Inspection first.** The external corpus undergoes the same automated inspection as the training corpus before any evaluation, covering structure, speaker identities, formats, sample rates, and durations. As Chapter 4 reports, this step uncovered a systematic association between sample rate and class that would otherwise have contaminated the result.

2. **Inclusion rule.** Only recordings of at least 30 seconds — the continuous read-speech tasks — are evaluated, because the short vowel and syllable tasks of that corpus do not match the continuous-speech conditions the pipeline is built for. The rule admits 216 recordings from all 61 speakers of the mirrored corpus.

3. **Bandwidth harmonization.** Every included file is first downsampled to 16 kHz — removing all content above 8 kHz for every speaker equally — and then resampled to the pipeline's standard 44100 Hz, after which the standard pipeline runs unchanged. This neutralizes the sample-rate confound identified during inspection.

4. **Frozen model.** The classifier trained on MDVR-KCL is applied without any retraining, threshold adjustment, or adaptation. Speaker-level scores are formed exactly as in Section 3.8 (mean over the speaker's recordings, threshold 0.5).

5. **Age-fair comparison.** The headline result compares elderly healthy controls against the (elderly) PD group; the young control group is reported separately, since comparing young controls with elderly patients would confound age with disease.

6. **Uncertainty reporting.** The fraction of results falling in the prototype's inconclusive band is reported alongside the metrics.

3.11 Prototype Design

3.11.1 Interfaces and Shared Prediction Path

The prototype offers two interfaces: a browser application (built with Streamlit [23]) supporting file upload and microphone recording, and a command-line tool suited to scripted use. Both call the same prediction function, which executes the pipeline of

Figure 3.1; the interfaces therefore cannot diverge in behaviour, and the automated tests of the prediction function cover both.

Before every prediction, the saved pipeline contract (Section 3.2) is verified. The input file is then validated with plain-language rejection messages for each failure category: missing file, empty file, undecodable audio, audio without samples, recordings shorter than one second, and silent recordings.

3.11.2 Score Interpretation and the Inconclusive Band

The classifier's PD score is not presented as a bare class label. Scores in the interval $[0.35, 0.65]$ are reported as inconclusive: "the research model could not clearly assign this recording to either class". Scores outside the band are reported as "the acoustic pattern was classified by the research model as closer to the PD (or HC) class". Figure 3.3 shows the decision rule. The band exists because scores near the threshold carry weak evidence, and because out-of-domain recordings — different microphones, rooms, or languages — concentrate near the boundary, as the external validation confirms; a hard label in that region would overstate what the model knows. The band's placement matches the region where the out-of-fold score distributions of the two classes overlap (Chapter 4).

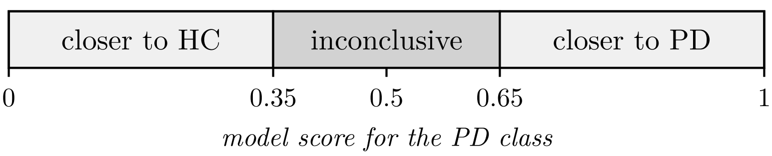

Figure (3.3): Interpretation of the classifier score in the prototype.

3.11.3 Recording-Condition Warnings

Non-blocking warnings accompany the result whenever the recording's conditions depart from those of the training corpus: a native sample rate different from 44100 Hz (the file is resampled, which can shift spectral features); audible clipping (more than 0.1% of samples at full scale); an estimated Signal-to-Noise Ratio (SNR) below 15 dB, computed as $20 \log_{10}$ of the ratio between the 90th and 10th percentiles of Root Mean Square (RMS) frame levels over 50 ms frames; and analysed durations under 10 seconds, far below the 70–220 s training recordings. Microphone mode carries an additional standing notice that live recordings differ from the training conditions and that the mode mainly demonstrates the system's handling of unfamiliar input.

3.11.4 Safety Wording and Data Handling

Every result, in both interfaces, is framed by the mandatory disclaimer: *"This is a research screening-support prototype and is not a medical diagnostic tool. Its result must not replace evaluation by a qualified healthcare professional."* The score is explicitly labelled a property of the research model rather than a person's medical risk, and the HC/PD labels are explained as similarity to dataset groups. Uploaded or recorded audio is written only to a temporary file that is deleted immediately after processing, even on failure; no audio is retained. Unexpected internal errors are never shown as technical traces: the user receives a short plain message, while diagnostic detail is written to the terminal log (and, in the command-line tool, exposed via a debug flag and distinct exit codes).

3.12 Development Environment and Tools

The system is implemented in Python 3.14 inside a project-local virtual environment. Praat [10], accessed through Parselmouth [11], computes the pitch, perturbation, HNR, and CPPS measures; librosa [12] computes the MFCC family and performs resampling and energy-based silence detection; NumPy [24] and SciPy [25] provide numerical routines; scikit-learn [16] provides the models, pipelines, and grouped splitters; Matplotlib [26] generates the result figures; and the pandas and SoundFile libraries handle tabular data and audio input/output.

Reproducibility measures include: fixed random seeds throughout; per-file caching of extracted features, so interrupted extraction runs resume rather than restart; one script per pipeline stage, each regenerating its report and artefacts from scratch; and version control of all code, configuration, reports, and documentation in a public repository (Appendix C). The audio corpora themselves are not redistributed — they are excluded from the repository in accordance with their licences, and download instructions are provided instead.

Chapter 4

Results and Discussion

4.1 Overview

This chapter reports the outcome of every stage of the methodology, in the order established in Chapter 3: dataset inspection (Section 4.2), feature extraction and validation (4.3), baseline modelling (4.4), the pre-declared model-selection study (4.5), the final model (4.6), its interpretation through feature importance (4.7), the empirical demonstration of the validation-design effect (4.8), external validation (4.9), and prototype verification (4.10). Section 4.11 discusses the findings as a whole, and Section 4.12 states the limitations and the ethical constraints that govern the system's use and wording. All classification results in this chapter were obtained under the subject-independent protocol of Section 3.8; the single deliberate exception, clearly marked, is the leaky split of Section 4.8, computed to demonstrate what the protocol prevents.

4.2 Dataset Inspection Results

The automated inspection of the MDVR-KCL corpus confirmed 73 readable recordings from 37 subjects (21 HC, 16 PD), organized in the structure of Table 3.1: all files mono WAV at 44100 Hz, with durations from 73.0 to 220.6 s (median 141.3 s, mean 141.4 s). Every integrity check passed: no corrupt, empty, silent, or very short files; no duplicate audio content by MD5 comparison; no label conflicts between folder and filename; no file without a parseable subject identifier; and no subject appearing in both classes. Subject-independent validation was therefore confirmed feasible, and no file had to be excluded.

The inspection also quantified the corpus' structural risks: 37 subjects is a small sample; the two recordings of each subject share one recording session and device; and the class ratio is 42:31 at the recording level. These findings directly motivated three methodological decisions already described — grouped validation, balanced class weighting with balanced accuracy as the primary metric, and the reporting of variability across folds and seeds.

4.3 Feature Extraction and Validation Results

Chunk-level extraction produced 1001 chunks (7–22 per recording) with all 74 features, with zero extraction failures. Automated validation of the feature tables found no infinite values, no constant (information-free) features, and no duplicated feature vectors. At the recording level the table contains no missing values; at the chunk level, a small number of chunks with too little stable voicing yielded missing perturbation values (at most 16 of 1001 chunks for any single feature), left as missing for fold-internal imputation as specified in Section 3.7.

One systematic observation required interpretation rather than correction: on some recordings the minimum of the F0 track lies marginally below the 75 Hz search floor (values down to about 74.94 Hz). This is normal behaviour of the pitch tracker, which interpolates between analysis frames [13]; the validation tolerances were set accordingly, and no data were altered.

As a physiological sanity check before any modelling, class means of key features were compared. The PD group showed higher mean local jitter (0.025 vs. 0.020), higher mean local shimmer (0.101 vs. 0.085), and slightly lower mean HNR (13.6 vs. 13.9 dB) than the HC group — the direction expected from the hypokinetic dysarthria literature reviewed in Section 2.3. The HC group also showed a higher mean F0 (about 180 vs. 154 Hz), a difference that could reflect the groups' sex composition rather than the disease; this concern is addressed quantitatively in Section 4.4.

4.4 Baseline Modelling Results

Table 4.1 reports the baseline phase: the 43-feature whole-recording representation (variant V0) evaluated for all five models. This phase used a single-seed instance of the protocol (seed 42, stratified grouped 5-fold splitting, metrics pooled over the out-of-fold predictions); the three-seed repetition was introduced with the model-selection study of Section 4.5. Every learned model clearly exceeds the majority-class baseline, whose balanced accuracy of exactly 0.5 — despite 0.575 plain accuracy — illustrates why balanced accuracy is the primary metric. The SVM performed best (subject-level balanced accuracy 0.780) and was provisionally selected at this stage by the robust-score rule (mean minus standard deviation of fold-level balanced accuracy).

Table (4.1): Baseline results (variant V0: 43 features per whole recording), subject-independent validation with a single seed (42), 5-fold, pooled over out-of-fold predictions. Balanced accuracy (bal. acc.) at recording and subject level; sensitivity, specificity, and AUC at recording level.

Model	Bal. acc. (recording)	Bal. acc. (subject)	Sensitivity	Specificity	AUC
Majority baseline	0.500	0.500	0.000	1.000	0.500
Logistic regression	0.651	0.662	0.516	0.786	0.717
SVM (RBF)	0.768	0.780	0.774	0.762	0.741
Random Forest	0.739	0.725	0.645	0.833	0.775
MLP	0.671	0.686	0.581	0.762	0.736

4.4.1 Confounder Check: Absolute Pitch and Speaker Sex

Because the corpus carries no verified per-subject sex metadata, and because the observed between-group F0 difference (Section 4.3) could reflect group composition, the entire evaluation was repeated with the four absolute-pitch features (mean, median, minimum, and maximum F0) removed. Table 4.2 shows the result. The SVM and logistic regression were unaffected or slightly improved — evidence that their decisions rest on the perturbation, cepstral, and spectral features rather than on absolute pitch. The MLP collapsed from 0.671 to 0.526, barely above chance, revealing that it had been relying substantially on the potentially sex-linked features; it was therefore excluded from all subsequent modelling.

Table (4.2): Ablation of the four absolute-F0 features (recording-level balanced accuracy, single seed 42, pooled out-of-fold; same protocol as Table 4.1).

Model	All features	Without absolute F0
Logistic regression	0.651	0.699
SVM (RBF)	0.768	0.780
Random Forest	0.739	0.707
MLP	0.671	0.526

4.5 Model-Selection Study

Table 4.3 reports all 19 configurations evaluated under the pre-declared protocol of Section 3.9 — every variant of Table 3.3 crossed with the remaining models, plus the tuned and ensemble variants. None reached the 0.95 investigation threshold. The V0 rows re-evaluate the baseline representation under the full three-seed protocol; their small differences from the single-seed values of Table 4.1 (for example, 0.704 versus 0.662 for logistic regression at subject level) reflect the averaging over additional seeds, not any change to the pipeline.

Applying the pre-declared rules, the adopted configuration is **V1 with the Random Forest**: 74 extended features extracted per 10-second chunk and averaged per recording, classified by a default-settings RF, reaching a subject-level balanced accuracy of 0.822 ± 0.032 — an improvement of +0.042 over the V0 baseline SVM. Three configurations with equal or higher point estimates were rejected because none cleared the +0.02 adoption margin over this simpler winner:

- the tuned-SVM+RF ensemble (V5), at 0.840, exceeds it by only +0.018 while adding nested tuning and a second model;
- the mean-and-standard-deviation summary with SVM (V6), at 0.824, exceeds it by +0.002 while doubling the feature count to 148;
- the tuned RF (V4), at 0.814, does not exceed the untuned RF at all.

These rejections are a designed outcome, not a missed opportunity: with seed-to-seed variability of about ±0.03 at this sample size, differences below the margin are indistinguishable from noise, and selecting the most complex of several statistically tied variants would fit the validation procedure rather than the problem. The gains that

Table (4.3): Complete results of the model-selection study. Primary metric in bold column: subject-level balanced accuracy, mean ± standard deviation over three seeds. AUC at subject level. LR = logistic regression; RF = Random Forest.

Variant	Model	**Bal. acc. (subject)**	Bal. acc. (recording)	AUC (subject)
V0	LR	0.704 ± 0.029	0.685	0.765
V0	SVM	0.780 ± 0.019	0.761	0.789
V0	RF	0.725 ± 0.019	0.725	0.808
V1	LR	0.746 ± 0.029	0.677	0.788
V1	SVM	0.816 ± 0.026	0.792	0.841
V1	**RF**	**0.822 ± 0.032**	**0.806**	**0.864**
V2	LR	0.769 ± 0.042	0.770	0.841
V2	SVM	0.817 ± 0.010	0.781	0.826
V2	RF	0.780 ± 0.011	0.750	0.802
V3	LR	0.772 ± 0.059	0.752	0.817
V3	SVM	0.793 ± 0.024	0.757	0.820
V3	RF	0.780 ± 0.011	0.775	0.820
V4	LR (tuned)	0.772 ± 0.028	0.751	0.829
V4	SVM (tuned)	0.775 ± 0.043	0.755	0.815
V4	RF (tuned)	0.814 ± 0.024	0.810	0.870
V5	SVM+RF ensemble	0.840 ± 0.026	0.800	0.853
V6	LR	0.656 ± 0.027	0.653	—
V6	SVM	0.824 ± 0.015	0.786	—
V6	RF	0.806 ± 0.023	0.794	—

survived the margin came from the data representation — the continuous-speech features and chunk averaging — while hyperparameter tuning contributed nothing, a pattern consistent with the small sample size.

4.6 Final Model Performance

Table 4.4 details the adopted configuration per seed. Averaged over seeds, the final model attains a subject-level balanced accuracy of 0.822 ± 0.032, sensitivity of 0.771, specificity of 0.873, and a subject-level AUC of 0.864; at recording level, balanced accuracy is 0.806 ± 0.013.

The pooled recording-level confusion matrix for seed 42 is: of 42 HC recordings, 36 correctly cleared and 6 falsely flagged; of 31 PD recordings, 24 correctly identified and 7 missed. In screening terms, the model catches roughly three of four PD cases while falsely flagging roughly one of eight healthy recordings.

Table (4.4): Final model (V1, Random Forest) per seed; all values subject-independent. Sensitivity, specificity, and AUC at subject level.

Seed	Bal. acc. (recording)	Bal. acc. (subject)	Sensitivity	Specificity	AUC
42	0.816	0.780	0.750	0.810	0.872
7	0.816	0.827	0.750	0.905	0.826
2025	0.788	0.859	0.812	0.905	0.893
Mean	0.806	0.822	0.771	0.873	0.864

Figure 4.1 shows the Receiver Operating Characteristic of the out-of-fold scores (seed 42) at both evaluation levels, with AUC 0.82 at recording level and 0.87 at subject level; subject-level aggregation improves ranking, as averaging two recordings suppresses single-recording noise. Figure 4.2 shows the out-of-fold score distributions by true class. The two classes separate visibly, but their overlap concentrates between roughly 0.35 and 0.65 — the empirical basis for the prototype's inconclusive band (Section 3.11.2).

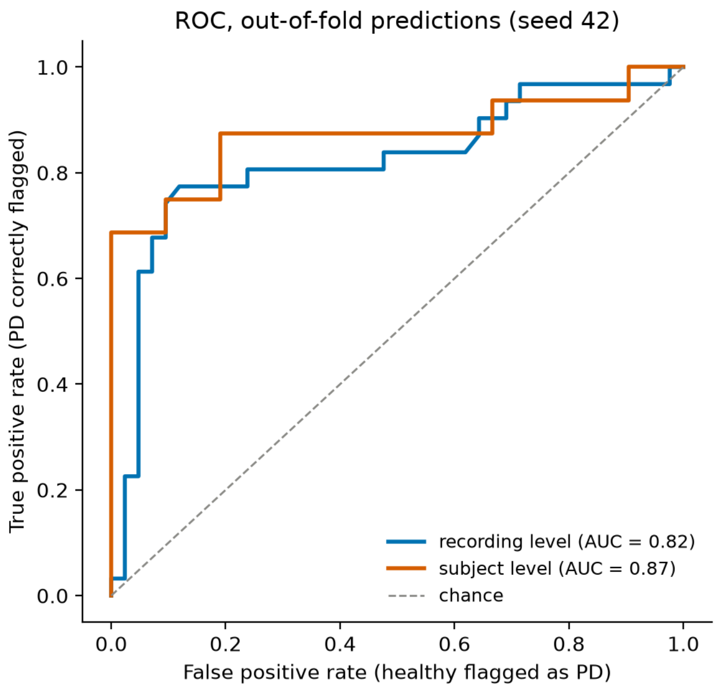

Figure (4.1): Receiver Operating Characteristic of the final model's out-of-fold predictions (seed 42), at recording and subject level.

Figure (4.2): Out-of-fold score distributions by true class (final model, seed 42, per recording). The shaded region marks the score range reported as inconclusive by the prototype.

4.7 Feature Importance and Interpretability

Figure 4.3 shows the fifteen highest-ranked features of the final Random Forest by mean decrease in impurity. The single most important feature is the F0 range (importance 0.069), followed by the F0 maximum (0.058) — direct acoustic correlates of the monotone, prosodically flattened speech that clinical descriptions identify as a hallmark of hypokinetic dysarthria [4, 8], and consistent with the discriminative power of F0 variability reported in early PD [5]. Variability measures of the spectral envelope and its dynamics (standard deviations of MFCC and delta-MFCC coefficients, e.g. 0.054 and 0.036 for the second MFCC and delta-MFCC) outrank the corresponding means, consistent with reduced articulatory movement rather than a shifted average spectrum. Absolute jitter (0.027) also appears among the leading features. The model's decisions therefore rest on acoustic quantities whose clinical interpretation is established — the explainability objective of Section 1.4.

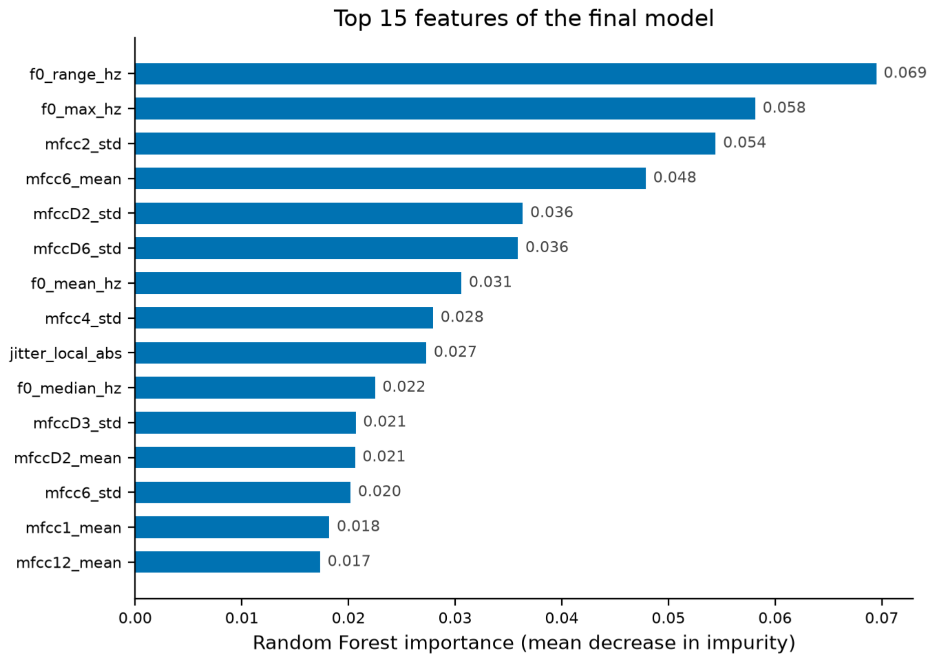

Figure (4.3): The fifteen most important features of the final Random Forest (mean decrease in impurity).

4.8 Effect of Validation Design: A Controlled Demonstration

To quantify the leakage mechanism discussed in Sections 2.6 and 3.8 on this project's own data, one controlled comparison was run: the identical chunk-level RF pipeline, with identical aggregation of chunk scores to recording level, evaluated twice — once with the correct subject-grouped split and once with a random split of the 10-second chunks, in which chunks of the same recording (and speaker) fall on both sides. Figure 4.4 shows the outcome: 0.775 balanced accuracy under the correct split against 0.909 under the wrong one. The +0.13 difference is produced entirely by the split; it measures the model's ability to recognize recordings and speakers it has partly seen, not the disease.

For completeness, the same comparison at the *recording* level showed almost no inflation on this corpus (0.807 under a random recording split against 0.806 grouped), because a subject's two recordings come from different tasks and are acoustically dissimilar. The severe leakage arises specifically when near-identical material — chunks of one recording — straddles the split, which is exactly the situation created by window-level splitting of segmented audio, a common design in the literature [19]. Reported accuracies above 0.9 on small corpora should therefore be read with this mechanism in mind, including this project's own leaky figure of 0.909, which is reported here only as a demonstration and is not a result of this work.

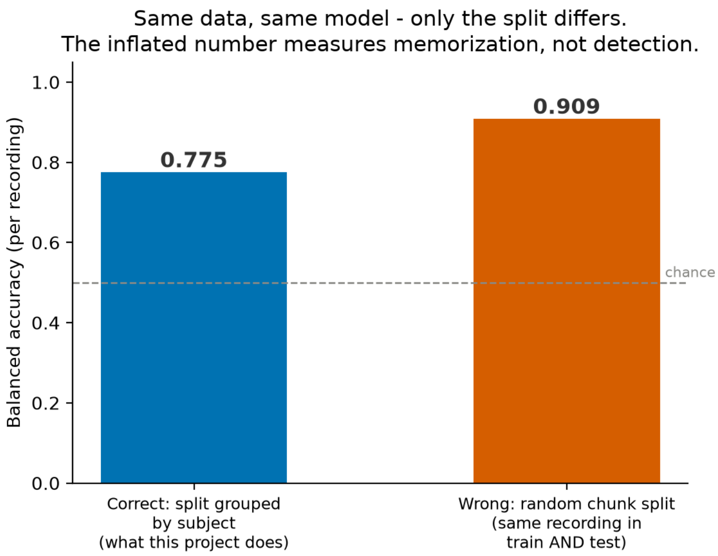

Figure (4.4): The same data, pipeline, and model under two split designs. The inflated figure measures memorization of recording identity, not detection (deliberately wrong split; demonstration only).

4.9 External Validation Results

4.9.1 Inspection and the Sample-Rate Confound

Inspection of the mirrored Italian corpus (Section 3.10) confirmed 831 readable mono WAV files from 61 speakers (15 young HC, 22 elderly HC, 24 PD) with zero corrupt files, and uncovered one critical structural fact: all 160 files sampled at 44100 Hz belong to the PD group, while every HC file is sampled at 16000 Hz. Since a 16 kHz recording cannot contain energy above 8 kHz while a 44.1 kHz recording can, a model using full-band spectral features could partly separate the groups by recording bandwidth alone — a channel artefact unrelated to any speaker's health. The bandwidth-harmonization step of Section 3.10 (downsampling every evaluated file to 16 kHz before the standard pipeline) was adopted for precisely this reason, and the result below would not be interpretable without it.

4.9.2 Results

Applying the inclusion rule admitted 216 recordings (≥30 s) from all 61 speakers, with zero extraction failures. Table 4.5 contrasts the frozen model's external performance with its internal performance. On the age-fair headline comparison (22 elderly HC vs. 24 PD speakers), the model attains a speaker-level balanced accuracy of 0.629 and an AUC of 0.701; 13 of 22 elderly HC speakers were correctly cleared (9 falsely flagged), and 16 of 24 PD speakers were caught (8 missed).

Table (4.5): Internal versus external performance of the identical frozen model (subject/speaker level). The external evaluation is cross-corpus and cross-language, with no adaptation.

Metric	Internal (MDVR-KCL)	External (Italian, elderly HC vs. PD)
Balanced accuracy	0.822	0.629
Sensitivity	0.771	0.667
Specificity	0.873	0.591
AUC	0.864	0.701

Two secondary observations carry as much information as the headline numbers. First, **the inconclusive band absorbed most of the out-of-domain material**: 71.3% of the 216 recordings scored inside $[0.35, 0.65]$, including the speaker-level scores of *all 22* elderly HC speakers (100%; PD speakers: 62.5%). Deployed as designed, the prototype would answer "inconclusive" rather than confidently mislabel foreign healthy speakers. Figure 4.5 shows every speaker's score.

Second, **the young healthy group scored anomalously high**: mean PD score 0.626, *above* the actual PD group's 0.566 (elderly HC: 0.489), with only 3 of 15 young speakers classified HC. Healthy young voices cannot plausibly be more parkinsonian than patients' voices; the anomaly demonstrates that under domain shift the score responds substantially to recording channel and speaking style rather than to disease. It is reported prominently here because it is direct empirical support for the cautious interface design — and a caution against interpreting any out-of-domain score clinically.

4.9.3 Interpretation

The external result quantifies generalization rather than assuming it. An AUC of 0.701 across corpus, language, microphone, and bandwidth — with zero adaptation — indicates

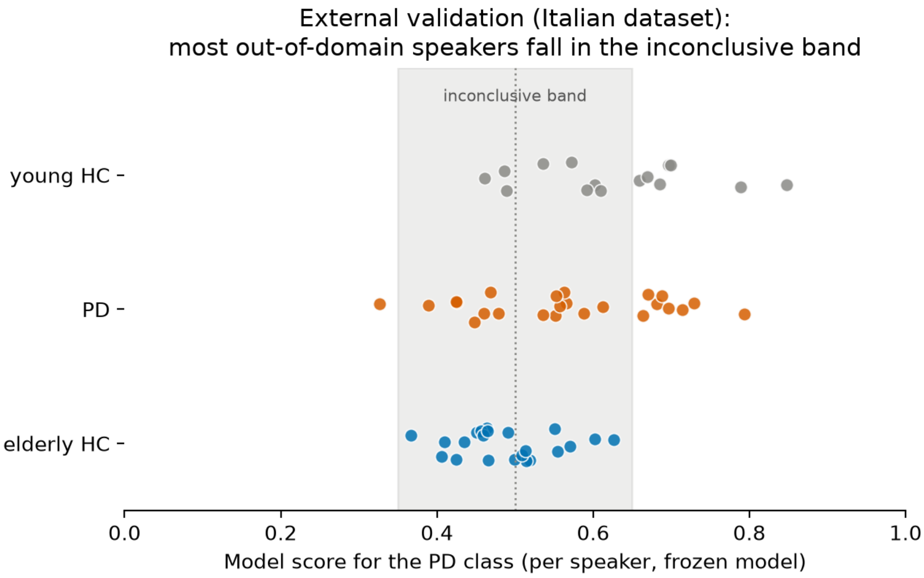

Figure (4.5): External validation: speaker-level scores of the frozen model on the Italian corpus, by group. The shaded region is the inconclusive band.

that a genuine, transferable acoustic signal survives: given one random PD and one random elderly-HC speaker, the model ranks the patient higher 70% of the time. At the same time, hard classification at the fixed 0.5 threshold degrades sharply (specificity 0.591), so the model is not usable as a classifier outside its training domain — which the prototype communicates by design. Because bandwidth harmonization removes spectral content the model saw in training, absolute scores shift, and the threshold-free AUC is the most meaningful external number. The gap between 0.822 and 0.629 is this project's measured price of domain shift; Chapter 5 returns to what would be required to reduce it.

4.10 Prototype Verification

The prototype was verified at three levels. First, automated tests of the shared prediction path all pass: correct end-to-end prediction on synthetic audio; refusal to predict under a modified, missing, or truncated pipeline contract; rejection, with plain-language messages, of missing, empty, corrupted, silent, and too-short audio files; and exact behaviour of the inconclusive band at its boundaries (scores of 0.349 and 0.651 classified, 0.35, 0.50, and 0.65 inconclusive). Second, the browser interface was exercised manually: both input modes render and complete analyses without server errors, and the microphone mode displays its out-of-domain notice. Third, an end-to-end command-line run on a corpus recording completed the full pipeline in about one minute for a 2.5-minute file — CPPS extraction dominating the runtime — and produced the following output (functional demonstration only; the file belongs to the training corpus, so the run verifies the pipeline, not accuracy):

```
================================================================
RESEARCH MODEL RESULT (non-diagnostic)
================================================================
File: data\raw\mdvr_kcl\ReadText\HC\ID00_hc_0_0_0.wav
Duration: 151.1 s (analyzed after trimming: 149.3 s),
original sample rate 44100 Hz

The acoustic pattern was classified by the research model as
closer to the HC class.
Model score for the PD class: 0.17 (0 = closer to HC,
1 = closer to PD).
This score is a property of the research model, NOT a
person's medical risk.
```

(The listing is wrapped to the page width; the disclaimer and label explanation that follow in the actual output are quoted in Section 3.11.4.)

4.11 General Discussion

Read together, the results support four statements.

First, *the acoustic signal is real but modest.* Under honest validation, continuous speech supports a subject-level balanced accuracy of about 0.82 on this corpus, and a ranking ability (AUC 0.70) that survives transfer to another language and recording channel. The features driving the result — F0 range, spectral variability, perturbation — correspond to the established clinical description of hypokinetic dysarthria, which strengthens the interpretation that the models measure the intended phenomenon.

Second, *validation design dominates headline numbers.* The same pipeline spans 0.775 to 0.909 depending only on the split (Section 4.8). This range exceeds the total improvement achieved by all feature engineering and model selection in this project combined (+0.042), which is the clearest argument that in this field the validation scheme deserves more scrutiny than the model.

Third, *improvements must be disciplined to be real.* Under pre-declared rules, most attempted improvements — tuning, ensembling, richer summaries — failed to clear a modest 0.02 margin. What remained (continuous-speech features and chunk averaging) is physiologically motivated and consistent across seeds. The 0.840 ensemble figure is reported, but adopting it on a 0.018 margin would have been selection noise.

Fourth, *uncertainty handling is a result, not a footnote.* The inconclusive band, placed where the classes' score distributions overlap, absorbed 100% of the elderly healthy speakers in the external test that the raw classifier would otherwise have misclassified at a rate near chance. For a screening-support concept, declining to answer out-of-domain is the correct behaviour, and the young-HC anomaly (Section 4.9) shows what confident answers would have looked like instead.

4.12 Limitations and Ethical Considerations

The findings above must be read within the following limitations, which also fix the ethical boundaries of the system's use.

4.12.1 Limitations of the Data

1. **Small sample.** 37 subjects; each validation fold tests 7–8 people, so single individuals move fold scores by several points. All results are therefore reported with their spread, and none should be quoted as a single number.

2. **One recording setup per corpus.** All training recordings share one device and environment; the model cannot distinguish "unhealthy voice" from "unfamiliar recording", as the external validation demonstrates.

3. **One language in training.** English only; the measured cross-language drop quantifies, but does not remove, this limitation.

4. **No demographic metadata.** Without verified per-subject sex and age, demographic confounding cannot be excluded — only bounded, as in the ablation of Section 4.4, which changed the selected model's result by less than 0.02.

5. **Class imbalance.** 42 HC versus 31 PD recordings, mitigated by balanced weighting and balanced accuracy but not removed.

6. **No severity information.** Early and advanced PD are one class; the system's sensitivity specifically to *early* disease — the motivating use case — is therefore unmeasured.

4.12.2 Limitations of the Method

1. Jitter, shimmer, and HNR are defined for sustained phonation and are noisier proxies on continuous speech; CPPS and the pause statistics partly, not fully, compensate.

2. Chunk-mean aggregation deliberately discards temporal detail in exchange for stability and explainability.

3. The classifier's score is not a calibrated probability; a score of 0.7 does not mean a 70% chance of disease, and the prototype's wording enforces this reading.

4. The system is cross-sectional: it compares a recording against dataset groups, never a person against their own history, although longitudinal change would likely be the more sensitive signal.

4.12.3 Limitations of the Evidence

1. One external corpus was tested; 0.701 AUC is a single data point about generalization, not a characterization of it.

2. Cross-validation characterizes performance over this population, not the reliability of any statement about an individual.

3. The 0.02 adoption margin, although pre-declared and consistently applied, is a judgement; a different reasonable margin would have selected the ensemble.

4.12.4 Ethical Constraints and Mandatory Wording

The system is a research screening-support prototype and is presented as such in every interface and in this report. Two concrete harms shape its wording. A *false reassurance* (a missed PD case — about one in four internally) could delay a genuine clinical visit; a *false alarm* (about one in eight internally, and roughly four in ten in the external setting) could cause real distress. The inconclusive band exists to reduce both in the uncertain region, and the mandatory framing addresses the rest: the output is always described as a non-diagnostic indication of similarity between acoustic patterns; the score is always labelled a property of the research model, not a person's medical risk; the HC/PD labels are explained as dataset-group similarity, explicitly not as the presence or absence of any condition; and every result carries the statement that it must not replace evaluation by a qualified healthcare professional. Formulations such as "diagnoses Parkinson's disease", "detects Parkinson's with certainty", "medical decision system", or "clinically validated" are prohibited and do not appear in the software or in this report. Finally, recordings submitted to the prototype are processed in a temporary file deleted immediately after analysis, no audio or result is stored or transmitted, and the audio corpora are used under their licences without redistribution.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This project set out to determine whether the acoustic pattern of a raw continuous-speech recording can be related to Parkinson's Disease status using reproducible signal processing and classical, explainable machine learning, evaluated honestly. Its findings are summarized in the following points.

- A complete pipeline was implemented from raw audio to a screening indication: five-step preprocessing, 10-second segmentation, extraction of 74 physiologically motivated acoustic features per chunk, chunk-mean aggregation, and classification — with one shared implementation for training, evaluation, and prediction, enforced by a saved configuration contract checked before every prediction.

- Under strictly subject-independent validation (stratified grouped 5-fold cross-validation, three seeds, with a runtime assertion excluding any subject overlap), the final model — a default-settings Random Forest on the 74 chunk-averaged features — achieved a subject-level balanced accuracy of 0.822 ± 0.032, sensitivity of 0.771, specificity of 0.873, and a subject-level AUC of 0.864 on the MDVR-KCL corpus (37 subjects).

- The model's decisions are interpretable and clinically coherent: the leading features are the fundamental-frequency range and maximum and the variability of the spectral envelope and its dynamics — acoustic correlates of the monotone, articulatorily reduced speech that defines hypokinetic dysarthria.

- Model selection was governed by rules declared before experimentation (identical splits, a +0.02 adoption margin, a 0.95 investigation threshold, full reporting of all 19 configurations). Under these rules, hyperparameter tuning, a model ensemble, and richer feature summaries were all rejected as within-noise; the genuine gains came from the continuous-speech features and chunk averaging (+0.042 over the baseline phase).

- The effect of validation design was demonstrated, not merely asserted: the identical pipeline reported 0.909 balanced accuracy under a deliberately wrong chunk-level split against 0.775 under the correct subject-grouped split — an inflation of +0.13 produced entirely by leakage, exceeding all genuine improvements of this project combined.

- External validation on an independent Italian corpus (61 speakers, different language, microphones, and bandwidth), with the model frozen and a bandwidth-harmonization step neutralizing a discovered sample-rate confound, yielded a speaker-level AUC of 0.701 and balanced accuracy of 0.629 on the age-fair comparison — evidence that a genuine, transferable acoustic signal survives domain shift, and a measurement of how much is lost to it.

- The screening prototype (browser application and command-line interface) reports an inconclusive result for scores in $[0.35, 0.65]$, attaches recording-condition warnings, retains no audio, and frames every output with mandatory non-diagnostic wording. In the external test, this design absorbed all elderly healthy speakers into the inconclusive band instead of confidently mislabelling them.

- All ten project objectives (Section 1.4) were met, and every reported number is regenerated by scripts in the public project repository (Appendix C).

The central conclusion is methodological as much as clinical: on small voice corpora, the choice of validation scheme moves reported performance more than model engineering does. An honestly validated 0.82, accompanied by a measured cross-corpus 0.70 AUC and an interface that declines to answer when uncertain, is a stronger scientific result than an inflated figure — and is the appropriate foundation for any future clinical development.

5.2 Future Work

The following directions, ordered roughly by expected value, would address the limitations stated in Section 4.12.

- **More and more diverse subjects.** The dominant limitation is 37 training subjects from one recording setup. Multi-centre corpora spanning devices, environments, and languages would both stabilize the estimates and let the model learn disease signal rather than channel signal.
- **Demographic metadata.** Verified per-subject sex and age would allow confounding to be controlled by design (matching or stratification) instead of only bounded by ablation.
- **Longitudinal, within-person analysis.** Comparing a person's recordings against their own history removes between-speaker variability entirely and is the more sensitive formulation for *early* detection; it requires repeated recordings over months or years.
- **Severity and stage labels.** Ratings such as UPDRS would enable regression toward disease severity and a direct test of sensitivity to early-stage cases — the motivating use case that binary labels cannot isolate.
- **Channel robustness and domain adaptation.** Techniques such as per-recording cepstral normalization, band-limited training, or calibration on a small target-domain sample could reduce the measured cross-corpus gap; each must be re-validated under the same subject-independent protocol.
- **Score calibration.** Mapping classifier scores to calibrated probabilities (with uncertainty) would make the inconclusive band statistically principled rather than empirically placed.
- **Broader external validation.** Additional public corpora in further languages would turn the single external data point into a generalization curve.
- **Clinical collaboration.** Any step beyond a research prototype — screening studies, deployment to real users — requires clinical partners, ethical approval, and regulatory framing that were deliberately outside this project's scope.

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Appendix A

Complete Feature Inventory

Table A.1 lists all 74 features in the exact order produced by the extraction stage (`src/pvoice/features.py`); this order is part of the saved pipeline contract (Section 3.2). The MFCC and delta-MFCC families are listed as ranges: each comprises the mean and standard deviation of coefficients 0–12.

Table (A.1): The complete 74-feature inventory.

Feature name	Meaning
Fundamental frequency statistics (7)	
`f0_mean_hz`	mean of the F0 track over voiced frames (Hz)
`f0_median_hz`	median F0 (Hz)
`f0_std_hz`	standard deviation of F0 (Hz); pitch variability
`f0_min_hz`	minimum tracked F0 (Hz)
`f0_max_hz`	maximum tracked F0 (Hz)
`f0_range_hz`	F0 maximum minus minimum (Hz)
`voiced_fraction`	fraction of frames with detected voicing
Jitter variants (4)	
`jitter_local`	mean absolute period-to-period difference, relative (Eq. 2.1)
`jitter_local_abs`	the same difference in seconds
`jitter_rap`	relative average perturbation (3-period)
`jitter_ppq5`	5-period perturbation quotient

Feature name	Meaning
Shimmer variants (5)	
shimmer_local	mean absolute amplitude difference, relative (Eq. 2.2)
shimmer_local_db	the same difference in decibels
shimmer_apq3	3-period amplitude perturbation quotient
shimmer_apq5	5-period amplitude perturbation quotient
shimmer_apq11	11-period amplitude perturbation quotient
Noise (1)	
hnr_mean_db	mean harmonics-to-noise ratio (dB, Eq. 2.3)
MFCC summary statistics (26)	
mfcc{0--12}_mean	mean of each of the 13 MFCCs over frames
mfcc{0--12}_std	standard deviation of each MFCC over frames
Cepstral (1)	
cpps_db	smoothed cepstral peak prominence (dB)
Pause and timing statistics (4)	
pause_count_per_min	pauses (≥0.2 s) per minute
pause_mean_s	mean pause duration (s)
pause_ratio	fraction of total time spent in pauses
speech_seg_mean_s	mean duration of continuous speech stretches (s)
Delta-MFCC summary statistics (26)	
mfccD{0--12}_mean	mean of each MFCC time-derivative over frames
mfccD{0--12}_std	standard deviation of each MFCC time-derivative

In the stored feature order, the base set of the baseline phase (43 features) comprises the F0, jitter, shimmer, and noise families followed by the MFCC summary statistics; the extended set appends the cepstral, pause, and delta-MFCC families.

Appendix B

Additional Experimental Results

This appendix supplements the result tables of Chapter 4 with the fold-level detail and confusion matrices of the baseline phase (single seed 42, stratified grouped 5-fold, as in Table 4.1). All values are taken from the generated report `reports/metrics_report.md` of the project repository.

B.1 Per-Fold Balanced Accuracy (Baseline Phase)

Table (B.1): Recording-level balanced accuracy of each fold.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Majority baseline	0.500	0.500	0.500	0.500	0.500
Logistic regression	0.533	0.812	0.312	0.833	0.708
SVM (RBF)	0.733	0.750	0.625	1.000	0.708
Random Forest	0.783	0.750	0.521	0.833	0.771
MLP	0.617	0.679	0.646	0.750	0.625

The fold-to-fold spread is large — for example, 0.312 to 0.833 for logistic regression, and a single perfect fold (1.000) for the SVM — which is the expected behaviour when each test fold contains only 7–8 subjects. This spread, and the seed-to-seed variability of about ±0.03 that it produces in the averaged results, form the empirical basis for the one-sided +0.02 adoption margin of the model-selection protocol (Section 3.9) and for reporting every headline number together with its variability.

B.2 Pooled Confusion Matrices (Baseline Phase)

Table B.2 shows the pooled out-of-fold confusion counts at both evaluation levels, in the form TN / FP / FN / TP (true HC classified HC / true HC classified PD / true PD classified HC / true PD classified PD). Recording level totals 42 HC and 31 PD recordings; subject level totals 21 HC and 16 PD subjects.

Table (B.2): Pooled confusion counts, baseline phase (TN/FP/FN/TP).

Model	Recording level	Subject level
Majority baseline	42/0/31/0	21/0/16/0
Logistic regression	33/9/15/16	16/5/7/9
SVM (RBF)	32/10/7/24	17/4/4/12
Random Forest	35/7/11/20	16/5/5/11
MLP	32/10/13/18	17/4/7/9

B.3 Final Model: Recording-Level Confusion by Seed

For the adopted configuration (V1, Random Forest), the pooled recording-level confusion for seed 42 is 36/6/7/24 (TN/FP/FN/TP), as reported in Section 4.6; the corresponding out-of-fold prediction tables for all three seeds, including every recording's score, subject identifier, task, and fold assignment, are stored in the repository (`data/processed/final_oof_predictions.csv` for seed 42, and regenerable for the others via `scripts/train_final_model.py`).

Appendix C

Software Repository and Reproduction Guide

All code, configuration, tests, generated reports, figures, and documentation of this project are version-controlled in a public repository:

`https://github.com/suhayl-alhammali/parkinsons_voice_project`

The audio corpora are *not* redistributed (they remain under their own licences); the repository contains download instructions instead. Every quantitative result in this report is regenerated by the scripts below.

C.1 Repository Structure

Path	Contents
`src/pvoice/`	the reusable pipeline package: configuration, dataset discovery, preprocessing and segmentation, feature extraction, models, grouped validation, prediction
`scripts/`	one script per pipeline stage (see below)
`tests/`	automated tests of the pipeline and prediction path
`app.py`	the browser prototype

`models/`	trained model and the pipeline-contract file
`reports/`	generated reports and figures
`docs/`	extended project documentation (12 documents)
`report/`	the LaTeX source of this report
`data/raw/`, `data/processed/`	corpora (user-supplied) and generated feature tables

C.2 Reproduction Sequence

All commands are executed in the project folder with the project's Python virtual environment active. Approximate runtimes refer to an ordinary laptop.

1. `pip install -r requirements.txt` — install dependencies (once).
2. Place the extracted MDVR-KCL corpus under `data/raw/mdvr_kcl/` (download instructions in that folder).
3. `python tests/test_pipeline.py` — pipeline self-test on synthetic audio (seconds).
4. `python scripts/inspect_dataset.py` — dataset inspection report (seconds).
5. `python scripts/build_features.py` — baseline 43-feature table (~12 min).
6. `python scripts/validate_features.py` — feature quality report (seconds).
7. `python scripts/train_models.py` — baseline evaluation of Table 4.1 (~2 min).
8. `python scripts/build_segment_features.py` — chunk-level 74-feature table (~32 min).
9. `python scripts/run_experiments.py` — the 19-configuration study of Table 4.3 (~30 min).
10. `python scripts/train_final_model.py` — final model, its report, and artefacts (~1 min).

11. `python scripts/make_figures.py` — all result figures, including the leakage comparison (~2 min).

12. External corpus inspection and frozen-model evaluation (~75 min; requires the Italian corpus under `data/raw/italian_pvs/`):
    `python scripts/inspect_italian_dataset.py`
    `python scripts/evaluate_external.py`

13. `streamlit run app.py` or `python scripts/predict_file.py <file.wav>` — the prototype.

Long extraction stages cache per-file results (`data/processed/*_cache/`) and resume automatically if interrupted. Fixed random seeds make every training and evaluation run deterministic.

C.3 Artefacts

The trained model is stored as `models/model.joblib` beside its pipeline contract `models/pipeline_config.json`, which records every preprocessing, segmentation, and feature setting together with the complete ordered feature list; the prediction path refuses to run if this contract does not match the code (Section 3.2). Out-of-fold prediction tables (`data/processed/*.csv`) preserve, for every recording, its subject, task, fold, true label, and score, allowing independent recomputation of every metric in this report.

Appendix D

Key Code Excerpts

The following excerpts are reproduced verbatim from the repository and implement the decisions most central to the report's claims. (The subject-overlap assertion, equally central, is already listed verbatim in Section 3.8.)

D.1 Chunk-Mean Aggregation at Prediction Time

From `src/pvoice/predict.py`: a new recording is segmented and aggregated by exactly the training-time rule (Eq. 3.1), using the same extraction functions.

```
    # Same rule as training: split into chunks, extract features per chunk,
    # average each feature over chunks (config.AGGREGATION).
    chunks = segment_audio(audio)
    chunk_rows = [extract_features(chunk) for chunk in chunks]
    mean_features = pd.DataFrame(chunk_rows).mean(axis=0).to_dict()
    X = pd.DataFrame([mean_features], columns=feature_names())
```

D.2 The Final Model Pipeline

From `scripts/train_final_model.py`: the adopted classifier as a three-stage pipeline, whose imputation and scaling statistics are fitted only on the data passed to `fit` — i.e., on training folds during validation (Section 3.7).

```
def build_pipeline() -> Pipeline:
    return Pipeline([
```

```
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler()),
        ("clf", RandomForestClassifier(
            n_estimators=300, class_weight="balanced",
            random_state=config.RANDOM_STATE)),
    ])
```

D.3 Score Interpretation (Inconclusive Band)

From `src/pvoice/predict.py`: the mapping of the classifier score to the three-way outcome of Figure 3.3, with the band limits taken from the central configuration (`UNCERTAIN_LOW` = 0.35, `UNCERTAIN_HIGH` = 0.65).

```
def classify_score(pd_score: float | None, fallback_label: str) -> tuple[str, str]:
    """Map a model score to (display label, headline sentence).

    Scores inside the configured uncertain band become "UNCERTAIN" rather
    than a hard class call; scores outside it keep the class the model
    predicted.  When no score is available, the raw class prediction is
    used as-is.
    """
    if pd_score is None:
        label = fallback_label
    elif config.UNCERTAIN_LOW <= pd_score <= config.UNCERTAIN_HIGH:
        return ("UNCERTAIN",
                "The research model could not clearly assign this "
                "recording to either class (the score is in the "
                "inconclusive middle range).")
    else:
        label = "PD" if pd_score > config.UNCERTAIN_HIGH else "HC"
    return (label,
            "The acoustic pattern was classified by the research model as "
            f"closer to the {label} class.")
```